

Petrographic Observations of the Apollo Lunar Thin Section 15415-r2

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1 Introduction

The lunar rock 15415, named the *Genesis Rock*, was collected by Apollo 15 astronauts David Scott and James Irwin in 1971 from the rim of Spur Crater on the lower slopes of Hadley Delta. The paired rotational series of thin section 15415-r2, initially downloaded from Virtual Microscope [1] and subsequently aligned by using Python in order to have consistent field of view in PPL and XPL.

2 Petrographic Observations

2.1 Observations in PPL

In PPL, the field is strikingly monotonous: virtually the entire view consists of a colorless, low-relief phase. There is no color contrast seen in the thin section. Large interlocking crystals occupy nearly the entire field of view. Grain boundaries are irregular and commonly sutured. Numerous fractures crosscut the crystals, producing a pervasive network of cracks throughout the section. Several broader fracture zones cutting diagonally across the field. Many cracks anastomose and branch, and the fracturing is pervasive rather than confined to grain margins. This degree of internal damage in an otherwise clear crystal is a first hint of shock deformation. No significant pleochroism is observed. The section is essentially a single mineral.

Scattered through the section are small rounded to elliptical inclusions, tens of micrometers to a fraction of a millimeter across, with higher relief and faintly darker rims. A single larger opaque or near-opaque grain with an irregular outline sits near the upper center of the field. These minor phases occupy only a percent or two of the area at most.

2.2 Observations in XPL

In XPL, the section displays predominantly low first-order gray and white interference colors. Individual grains exhibit distinct extinction positions, producing a mosaic texture of adjacent crystals with different optical orientations. Many grains display prominent polysynthetic (albite) twinning, visible as closely spaced alternating light and dark lamellae. Fractures visible in PPL remain evident in XPL and locally offset or truncate twin lamellae. The rock appears coarse-grained, with individual crystals extending across substantial portions of the field of view. The field is likely

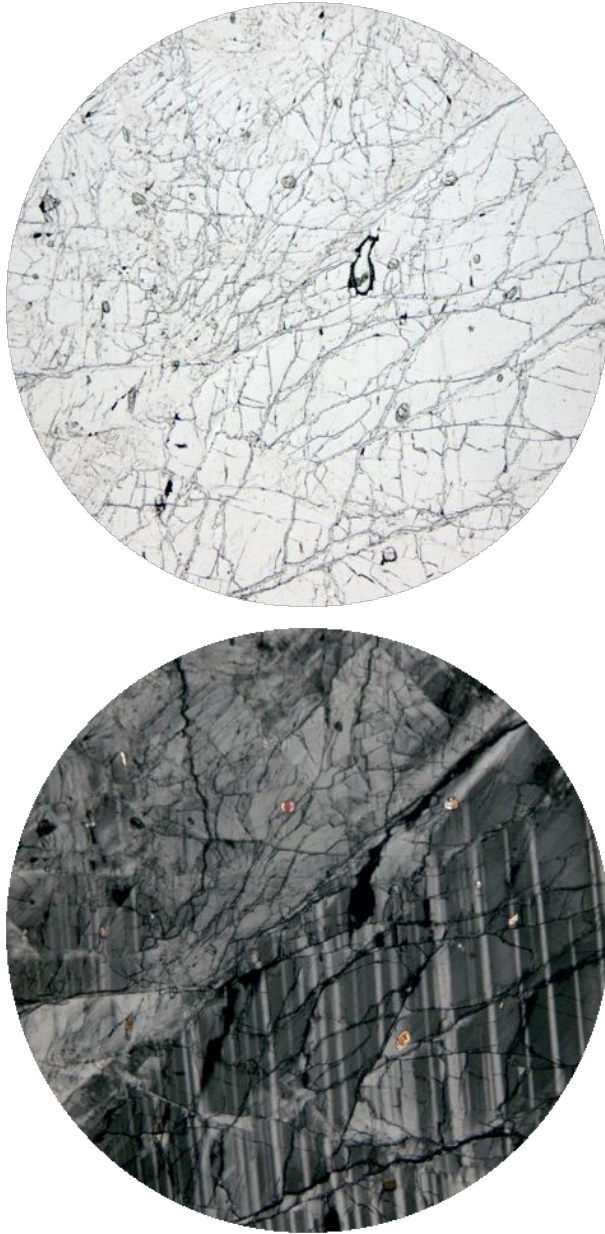


Figure 1: Sample 15415-r2 in PPL (top) and XPL (bottom). Field of view 2.0 mm



Figure 2: Sample 15415 under reflected light

occupied by very large plagioclase crystals. Several observations stand out in XPL:

Grain size and boundaries. Only a few crystals occupy the entire field of view. A boundary can be traced diagonally across the image where the twin lamellae abruptly change orientation, separating one large twinned individual from another. At the magnification of these images, single crystals extend beyond the circular field — the section is effectively a mosaic of a handful of giant plagioclase grains.

Twinning. The albite twinning appears slightly bent or tapered, and some domains show patchy, undulose extinction where the gray tones sweep irregularly across the lamellae rather than extinguishing uniformly. Bent twin lamellae and undulatory extinction in plagioclase are classic indicators of mild shock and strain — entirely expected for a rock that has resided at or near the lunar surface, surviving fragmentation events during its long exposure history.

Fractures. The crack networks seen in PPL remain prominent in XPL as dark seams crosscutting the twin lamellae. Because the fractures cut across the twinning rather than following it, they clearly postdate crystal growth — mechanical damage superimposed on an originally coherent igneous crystal.

Inclusions. The small rounded grains noted in PPL now reveal low- to moderate-birefringence interference colors, several showing warm yellow-orange to pinkish tints. These are consistent with the minor pyroxene known from this sample, which occurs as small blebs within and between the plagioclase grains, with trace ilmenite accounting for the opaque grain. Published modal analyses of 15415 report roughly 97–99% plagioclase with about 1–3% pyroxene and a trace of opaques, in good agreement with what is visible here [2–5].

2.3 Texture

The sample exhibits a coarse-grained, holocrystalline texture dominated by plagioclase. The interlocking grain boundaries and abundance of plagioclase indicate an anorthositic composition. Extensive fracturing and local brecciation textures suggest that the rock has experienced significant impact-related deformation after crystallization. Despite this deformation, the primary igneous plagioclase fabric remains largely preserved.

2.4 Interpretation

The dominance of coarse plagioclase, together with the scarcity of mafic minerals, is characteristic of a ferroan anorthosite. Such rocks are interpreted as fragments of the primordial lunar highland crust. The abundance of plagioclase supports the Lunar Magma Ocean model, in which buoyant anorthitic plagioclase crystallized and floated to the surface of a global magma ocean, forming the early lunar crust. The pervasive fractures likely record subsequent impact events that affected the lunar highlands.

3 Segmentation of Opaques

You may want to give it a try to segment opaque phases in the PPL and XPL images, allowing the texture of the sample to be visualized more clearly: <https://huggingface.co/spaces/pingwang2026/segmentation-of-opaques-in-petrographic-images>

References

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